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Decision Tree Algorithm

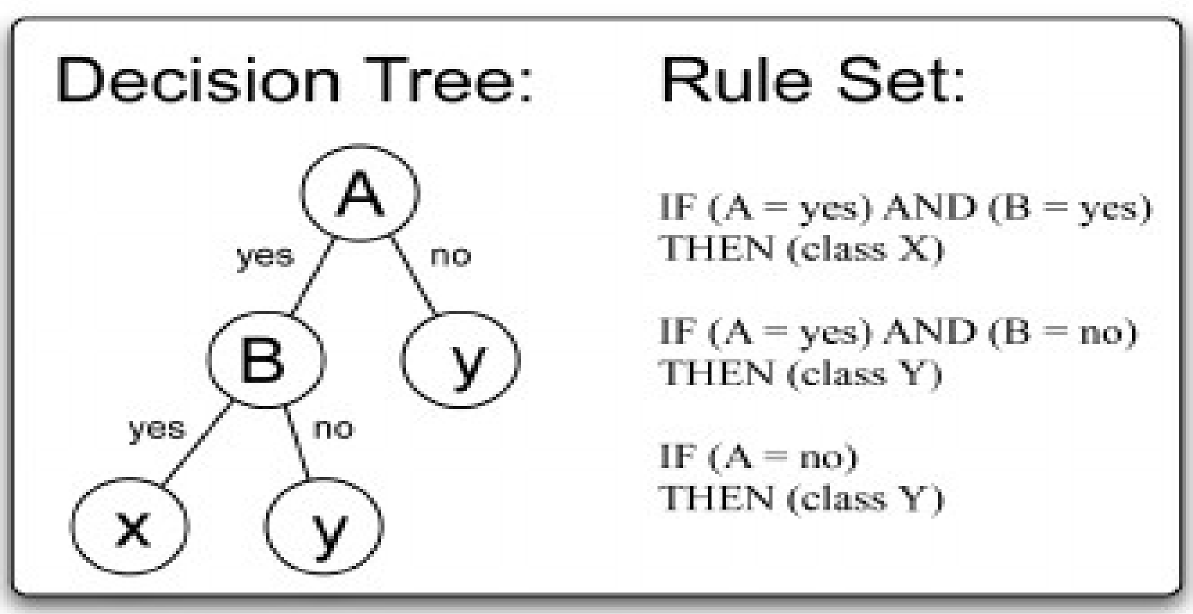
From classrooms to corporate, most important is decision trees.

Mathematics behind decision tree is very easy to understand compared to other machine learning algorithms.

Decision tree is also easy to interpret and understand compared to other ML algorithms.

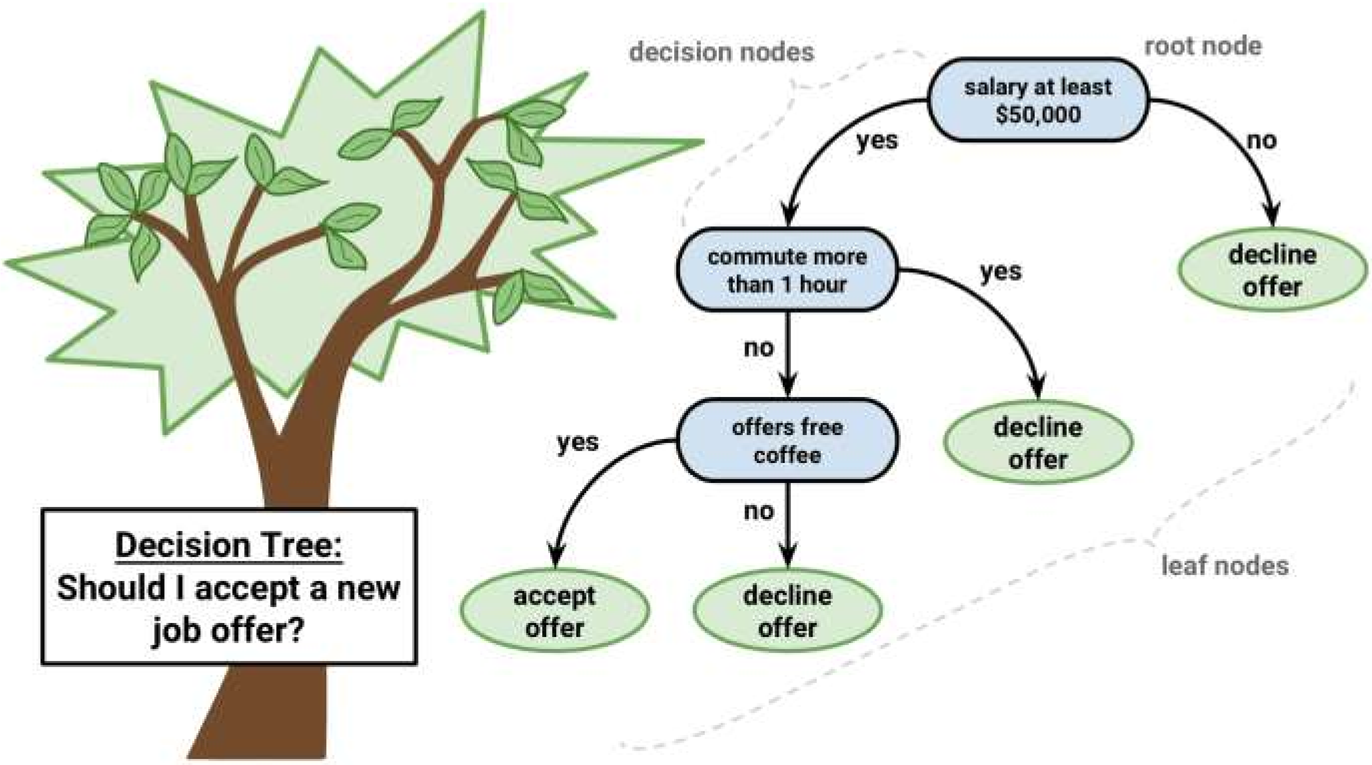
→ **Non-parametric** Supervised Machine Learning Algorithm

→Uses a set of rules to make decisions, similarly to how humans make decisions. **Nested if-else condition .**



→ A rule is a conditional statement that can easily be understood by humans and easily used within a database to identify a set of records.

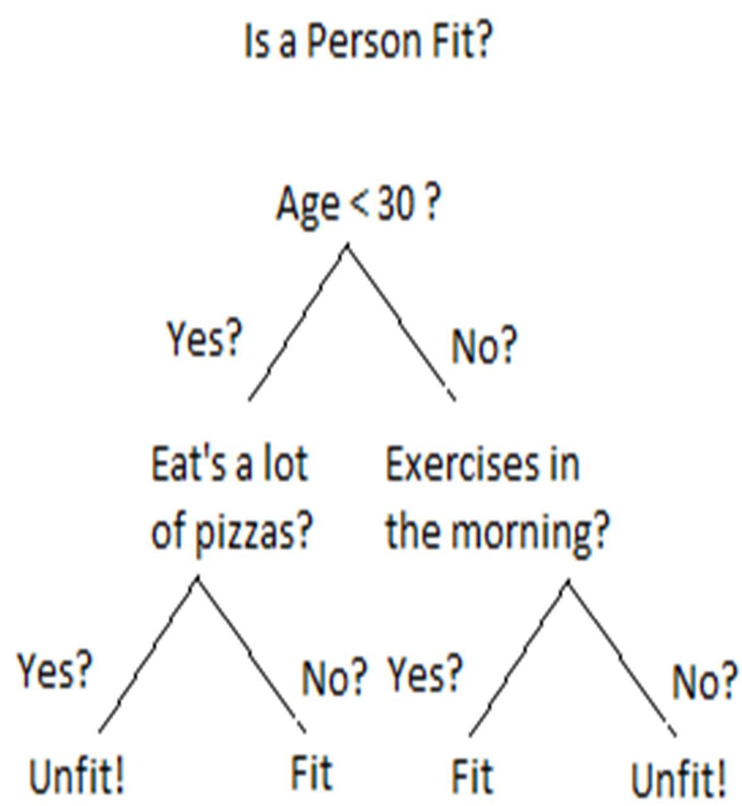
Decision trees can perform both classification and regression tasks.



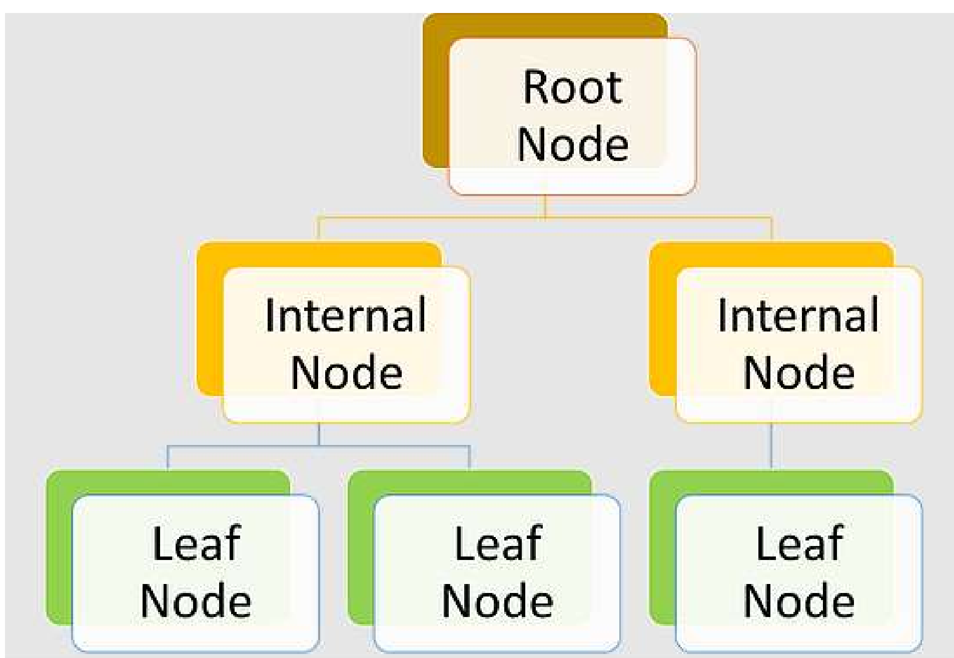
Decision tree based on the nested if-else classifier.
In general, a decision tree takes a statement or hypothesis or condition and then makes a decision on whether the condition holds or does not.
The conditions are shown along the branches and the outcome of the condition, as applied to the target variable.

Terminology:

1. **Root Node:** It represents entire population or sample and this further gets divided into two or more homogeneous sets.
2. **Splitting:** It is a process of dividing a node into two or more sub-nodes.
3. **Decision Node:** When a sub-node splits into further sub-nodes, then it is called decision node. This ultimately leads to a prediction.
4. **Leaf/ Terminal Node:** Nodes (no further split) is called Leaf or Terminal node.
5. **Pruning:** When we reduce the size of decision trees by removing nodes (opposite of Splitting), the process is called pruning.
6. **Branch / Sub-Tree:** A sub section of decision tree is called branch or sub-tree.



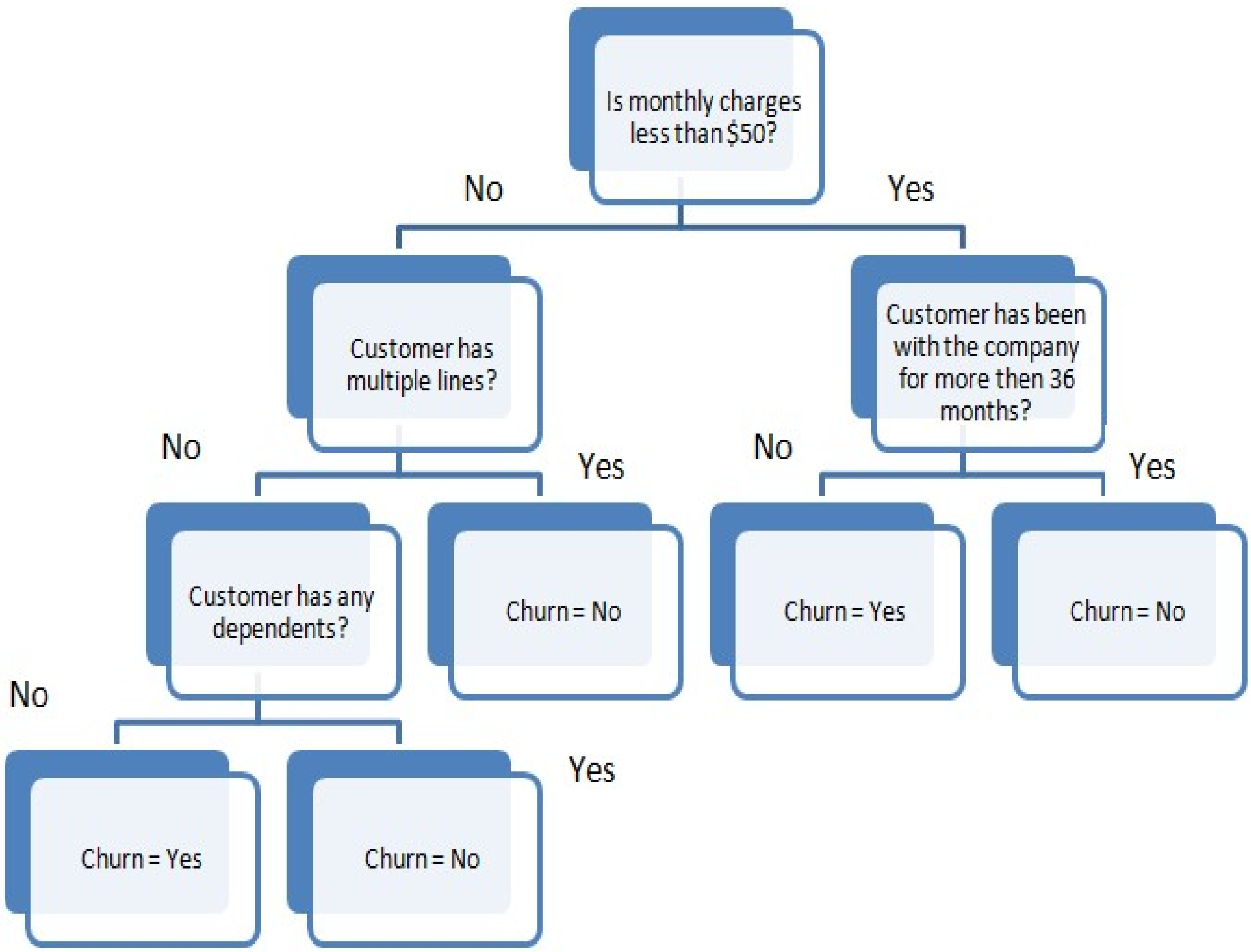
you start question from root node then move through the tree branches according to which groups you belong to until you reach a leaf node.



Variable selection criterion

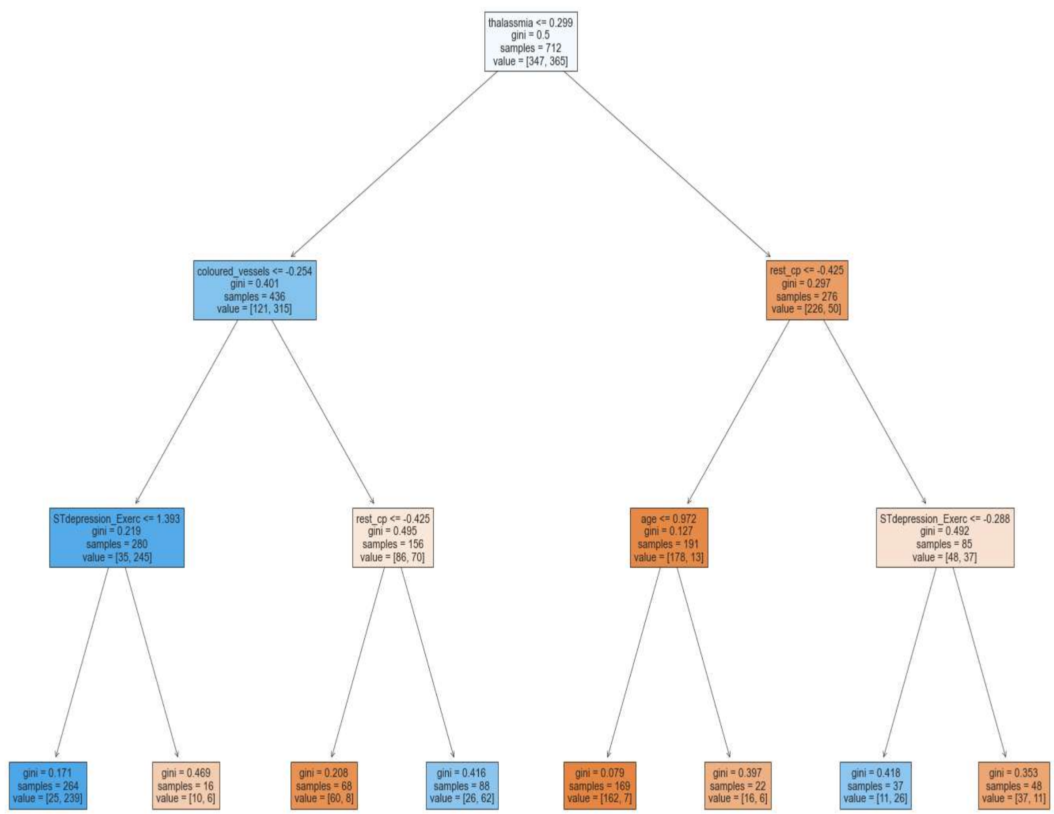
Example of telecommunication churning customers.:-

DECISION TREE CLASSIFIER:-



Mathematics behind the calculation to decide which variable should be ROOT Node , next. Node .. and

#	Column	Non-Null Count	Dtype
0	age	1025 non-null	int64
1	gender	1025 non-null	int64
2	rest_cp	1025 non-null	int64
3	rest_bp	1025 non-null	int64
4	cholesterol	1025 non-null	int64
5	fast_bloodsugar	1025 non-null	int64
6	rest_ecg	1025 non-null	int64
7	stress_HR	1025 non-null	int64
8	Exercise_cp	1025 non-null	int64
9	STdepression_Exerc	1025 non-null	float64
10	STpeak_exerc	1025 non-null	int64
11	coloured_vessels	1025 non-null	int64
12	thalassmia	1025 non-null	int64
13	heart_disease	1025 non-null	int64



Mathematics behind Decision Tree:

1. Flow of information through the Decision Tree

2. How does Decision Trees select which variable to split on at decision nodes?

3. How does it decide that the tree has enough branches and that it should stop splitting?

Variables are selected on a statistical criterion which is applied at each decision node.

variable selection criterion in Decision Trees can be done via two approaches: -

1. Entropy and Information Gain
2. Gini Index

Both criteria are broadly similar and seek to determine which variable would split the data to lead to the underlying child nodes being most homogenous or pure.

Entropy

Entropy is amount of information is needed to accurately describe some sample. So if sample is homogeneous, means all the element are similar than Entropy is 0, else if sample is equally divided than entropy is maximum 1.

Mathematically it is written as ,

$$E = -\sum_{i=1}^{n} p_i \, log_2(p_i)$$

Where the pi is the probability of randomly selecting an example in class i.

Gini index / Gini impurity

Gini index is measure of inequality in sample. It has value between 0 and 1. Gini index of value 0 means sample are perfectly homogeneous and all element are similar, whereas, Gini index of value 1 means maximal inequality among elements. It is sum of the square of the probabilities of each class.
It is illustrated as,

$$Gini\,index = 1 - \sum_{i=1}^{n} p_i^2$$

i is number of classes

The Entropy and Information Gain method focuses on purity and impurity in a node.

The Gini Index or Impurity measures the probability for a random instance being misclassified when chosen randomly.

Solving problem prediction of student pass fail in exam Calculation of Gini Index: -

The dataset of student pass fail in exam:

Resp srl no	Target variable	Predictor variable	Predictor variable	Predictor variable
	Exam Result	Other online courses	Student backgroun d	Working Status
1	Pass	Y	Maths	NW
2	Fail	N	Maths	W
3	Fail	y	Maths	W
4	Pass	Y	CS	NW
5	Fail	N	Other	W
6	Fail	Y	Other	W
7	Pass	Y	Maths	NW
8	Pass	Y	CS	NW
9	Pass	n	Maths	W
10	Pass	n	CS	W
11	Pass	y	CS	W
12	Pass	n	Maths	NW
13	Fail	y	Other	W
14	Fail	n	Other	NW
15	Fail	n	Maths	W

Gini Index

The Gini Index or Impurity measures the probability for a random instance being misclassified when chosen randomly.

The lower the Gini Index, the better the lower the likelihood of misclassification.

The formula for Gini Index

$$Gini = 1 - \sum_{i=1}^{C} (p_i)^2$$

P(i) represents the ratio of Pass/Total no. of observations in node.

Calculate the Gini index for Student Background attribute.

There are three Sub nodes :- Maths, CS, Others.

Gini formula requires us to calculate each sub node.

Then do a weighted average to calculate the overall Gini Index for the Background node.

Maths sub node: 4 Pass, 3 Fail

$$Gini_{maths} = 1 - \left(\frac{4}{7}\right)^2 - \left(\frac{3}{7}\right)^2 = .4897$$

CS sub node: 4Pass, 0 Fail

$$Gini_{CS} = 1 - \left(\frac{4}{4}\right)^2 - \left(\frac{0}{5}\right)^2 = 0$$

Others sub node: 0Pass, 4 Fail

$$Gini_{others} = 1 - \left(\frac{0}{4}\right)^2 - \left(\frac{4}{4}\right)^2 = 0$$

The overall Gini Index for Background :-

$$Gini_{bkgrd} = \frac{7}{15} * .4897 + \frac{4}{15} * 0 + \frac{4}{15} * 0 = \ .2286$$

Gini Index for Working Status :-

$$Gini_{working} = 1 - \left(\frac{6}{9}\right)^2 - \left(\frac{3}{9}\right)^2 = .44$$

$$Gini_{notworking} = 1 - \left(\frac{5}{6}\right)^2 - (6)^2 = .278$$

$$Gini_{workstatus} = \frac{9}{15} * .44 + \frac{6}{15} . 278 = .378$$

Gini Index for Other Online Courses.

$$Gini_{online} = 1 - \left(\frac{5}{8}\right)^2 - \left(\frac{3}{8}\right)^2 = .4688$$

$$Gini_{notonline} = 1 - \left(\frac{3}{7}\right)^2 - \left(\frac{4}{7}\right)^2 = .4898$$

$$Gini_{\text{Other Online}} = \frac{8}{15} * .4688 + \frac{7}{15} . 4898 = .479$$

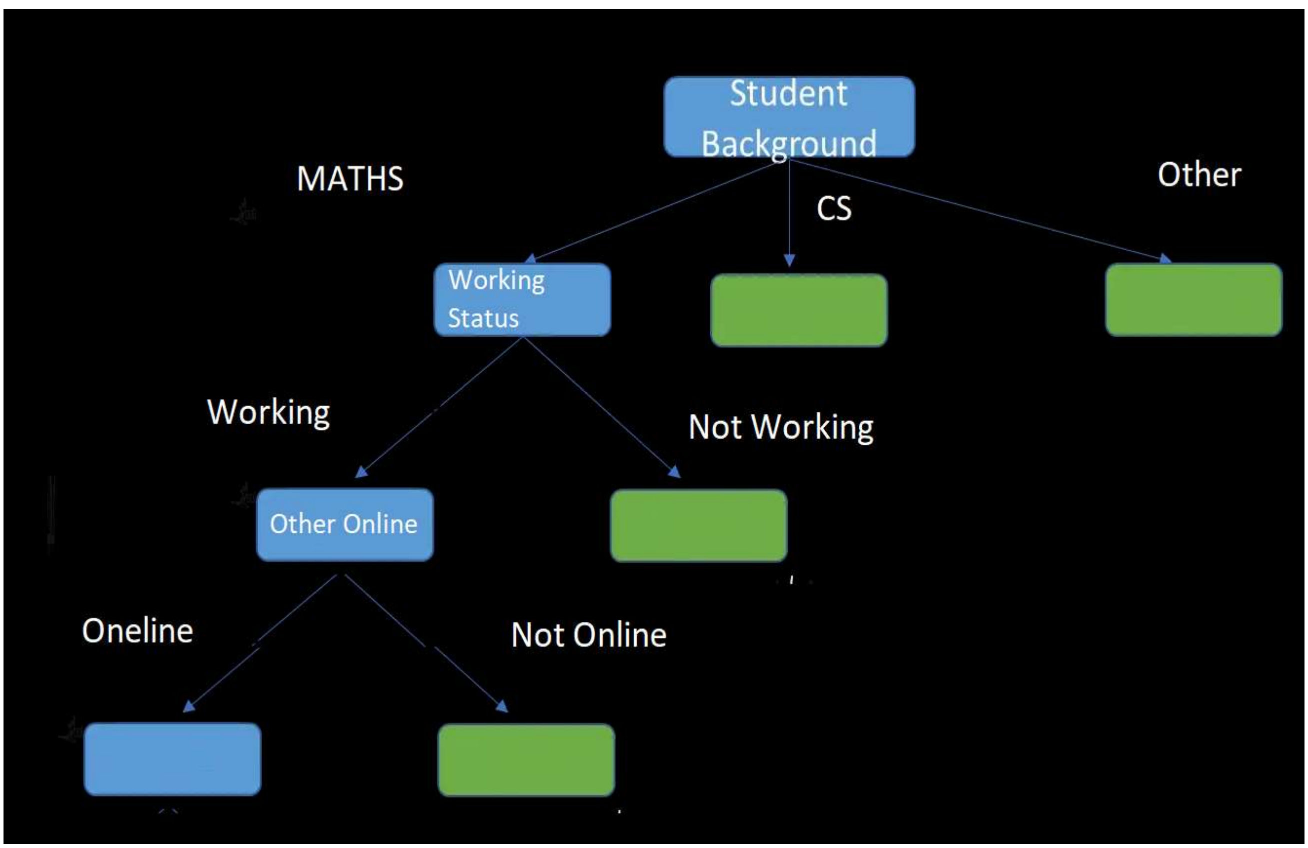
Final Gini Values of different Nodes:-

$$Gini_{bkgrd} = \frac{7}{15} * .4897 + \frac{4}{15} * 0 + \frac{4}{15} * 0 = .2286$$

$$Gini_{workstatus} = \frac{9}{15} * .44 + \frac{6}{15} . 278 = .378$$

$$Gini_{\textbf{Other Online}} = \frac{8}{15} * .4688 + \frac{7}{15} . 4898 = .479$$

The Gini Index is lowest for the Student Background variable.



```
from sklearn.tree import DecisionTreeClassifier# classification
#from sklearn.tree import DecisionTreeRegressor# Regression
dt=DecisionTreeClassifier(criterion="entropy")
# criterion=entropy/gini
dt.fit(X_train1,Y_train1)
Y_pred_dt_train=dt.predict(X_train1)
Y_pred_dt_test=dt.predict(X_test1)
Print(accuracy_score(Y_train1,Y_pred_dt_train))
print("######"*20)
print(accuracy_score(Y_test1,Y_pred_dt_test))
```

Overfitting and Decision Trees

Overfitting can be a big challenge with Decision Trees.

There are several ways we can prevent the decision tree from becoming too unwieldy: -

1. Pre pruning or Early stopping: Preventing the tree from growing too big or deep
2. Post Pruning: Allowing a tree to grow to its full depth and then getting rid of various branches based on various criteria

3. Ensembling or using averages of multiple models such as Random Forest

Pruning

Pre pruning

The pre-pruning technique refers to the early stopping of the growth of the decision tree. The pre-pruning technique involves tuning the hyperparameters of the decision tree model prior to the training.

The hyperparameters

max_depth, min_samples_leaf, min_samples_split

We can set max_depth or min_samples parameters in order not to overcomplicate the tree.

which can be tuned to early stop the growth of the tree and prevent the model from overfitting.

The best way is to use the RandomSearchCV or GridSearchCV technique to find the best set of hyperparameters.

A challenge with the early stopping may prevent some more fruitful splits down the line.

Post Pruning

In this technique, we allow the tree to grow to its maximum depth. Then we remove parts of the tree to prevent overfitting.

A good method to avoid overfitting is pruning — it removes insignificant nodes.

Advantages and Disadvantages of Trees
Decision trees

1. Trees give a visual schema of the relationship More explainable. The hierarchy of the tree provides insight into variable importance.

2. White box model which is explainable. This is in contrast to black box models such as neural networks.

Disadvantages

1. Prone to overfitting and hence lower predictive accuracy

2. Can be non-robust, i.e., a small change in the data can cause a large change in the final estimated tree

3. Decision tree learners create biased trees if some classes dominate. It is required to balance the dataset prior to fitting with the decision tree.